

Total No. of Questions : 10]

SEAT No. :

P976

[Total No. of Pages : 3

[5871]-753

B.E. (Honours)

ARTIFICIAL INTELLIGENCE AND MACHINE

Soft Computing and Deep Learning

(2015 Pattern) (Semester - II) (410303)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8. Q. 9 or Q. 10.
- 2) Assume suitable data, if necessary.
- 3) Neat diagrams must be drawn whenever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Differentiate Supervised Learning and Unsupervised Learning. [3]

b) What is soft computing? Indicate biological analogies of basic techniques of soft computing. Describe why soft computing is particularly useful in representing and reasoning with human-oriented knowledge. [6]

OR

Q2) a) Define the following for membership functions: [3]

- i) Core
- ii) Support
- iii) Boundary

b) Define Crisp and Fuzzy relation. Describe fuzzy max-min composition, fuzzy min-max composition and max-product composition. [6]

Q3) a) Explain in detail Mamdani FIS.(Fuzzy Inference System) [3]

b) Explain fuzzification and defuzzification techniques. [6]

OR

P.T.O.

- Q4)** a) Explain in detail convergence working principle in genetic algorithm. [3]
b) What is genetic programming? Write and explain the executional steps of Genetic Programming. [6]

- Q5)** a) Explain working of Bayesian Neural Networks with suitable diagram. [6]
b) What is Neural Network? Explain with suitable diagram the different classes of network architecture. [6]
c) Explain the following terms: [4]
i) Theano in Deep Learning.
ii) Caffe in Deep Learning.

OR

- Q6)** a) Explain in detail various Open Source Deep Learning Libraries and Platforms. [6]
b) What do you mean by Tensor in TensorFlow? What are the Programming elements of TensorFlow? [6]
c) Explain in detail Feed Forward Neural Network. [4]

- Q7)** a) Explain with example Self Organizing Maps. [8]
b) What is a Deep Belief Network? How did Deep Belief Neural Networks Evolve? [6]
c) What is image captioning in Deep Learning? [4]

OR

- Q8)** a) What is the role of the Activation functions in Neural Networks? Explain in detail some of the popular Activation Functions used in Neural Networks. [8]
b) What is pooling? What are the different types of Pooling? Explain their characteristics. [6]
c) What do you mean by Perceptron? Explain in detail different types of Perceptrons? [4]

- Q9) a)** Explain following with example [8]
- i) Sequential processing LSTM
 - ii) Recurrent Neural Network(RNN)
- b) Explain in detail architecture Convolution Neural Network (CNN). Write down various properties of CNN representations. [6]
- c) Explain on detail Architecture of Back Propagation (BP) Networks. [4]

OR

- Q10)a)** What is associate memory in neural network? Explain various types and applications of associative memory. [8]
- b) Write short notes on: [6]
- i) Artificial Neural Network (ANN)
 - ii) Application of RNN
- c) Explain Adaptive Resonance Theory? What is the major difference between ART1 and ART2? [4]

